Introduction to Software Engineering

COSC 336

ReSource (Phases IV and V)

Design Document

Architecture Design and Detailed Design

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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Sections in This File

This file contains the following parts of the ReSource Phase 4 and 5 Design Document. Section, figure and table numbers are the same as in the combined group document.

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- 7.2 Request Submission
- 7.4 Transfer Approval
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7. Mockups

The eight screens below are the ones listed in the project description. Each screen shows the role, real fields with sample data, the AI result with its confidence and reason, the decision the person takes and the state shown when the AI is unavailable. Numbered markers on each screen point to these five parts: (1) AI result, (2) confidence, (3) reason, (4) human decision and (5) AI-down state. Table 7-1 summarizes the screens.

Table 7-1: Mockup screens

Screen	Role	Use case	AI result shown	Human decision	AI-down state	Requirements
7.1 Asset Registration	Asset Coordinator, Electrical Engineering	UC-1, UC-10	Category, tags and description suggestions with confidence and reason	Accept, edit or reject each field	Fields stay empty; choose from the list	FR-1.2, FR-9.1 to FR-9.9
7.2 Request Submission	Department Representative, Physics	UC-3, UC-2	Existing listings that may meet the need, with fit score and reason	Request a listing or submit an open need	Standard search, results not ranked	FR-3.2, FR-3.3, FR-3.13, FR-8.1
7.3 Recommendation Review	Procurement Team	UC-9	Ranked matches, factor values, confidence, explanation	Select, choose another or reject with a reason	Keyword and rule matches, not ranked	FR-8.3 to FR-8.9, LEG-9
7.4 Transfer Approval	Administrator	UC-4	Match score and reasons that led to the request	Approve or reject with a reason	Approval works; AI details hidden	FR-4.2 to FR-4.6
7.5 Maintenance Tracking	Maintenance Staff	UC-6, UC-7	Sustainable-action suggestion with confidence and estimates	Record results; propose or reject the action	No suggestion; manual review	FR-5.4 to FR-5.12, FR-10
7.6 Sustainability Dashboard	Sustainability Officer	UC-12	AI highlights of large changes	Review indicators and request a report	Indicators shown, no highlights	FR-13.1 to FR-13.12, FR-12.8
7.7 AI Report Review	Finance Officer	UC-12	AI report draft with a figure check and assumptions	Approve, edit or discard	Standard report opens	FR-12.1 to FR-12.11, BR-11
7.8 Conversational Interaction	Department Representative	UC-11	Answer with linked listing and a request draft	Confirm, edit or cancel	Assistant unavailable; draft kept	FR-11.1 to FR-11.10

7.1 Asset Registration

The Asset Coordinator of Electrical Engineering registers the surplus oscilloscopes. The AI suggestions appear in a separate panel, none is pre-accepted, and each field is accepted, edited or rejected on its own (FR-9.6). The hazard flag is entered by the person, never suggested (FR-1.15).

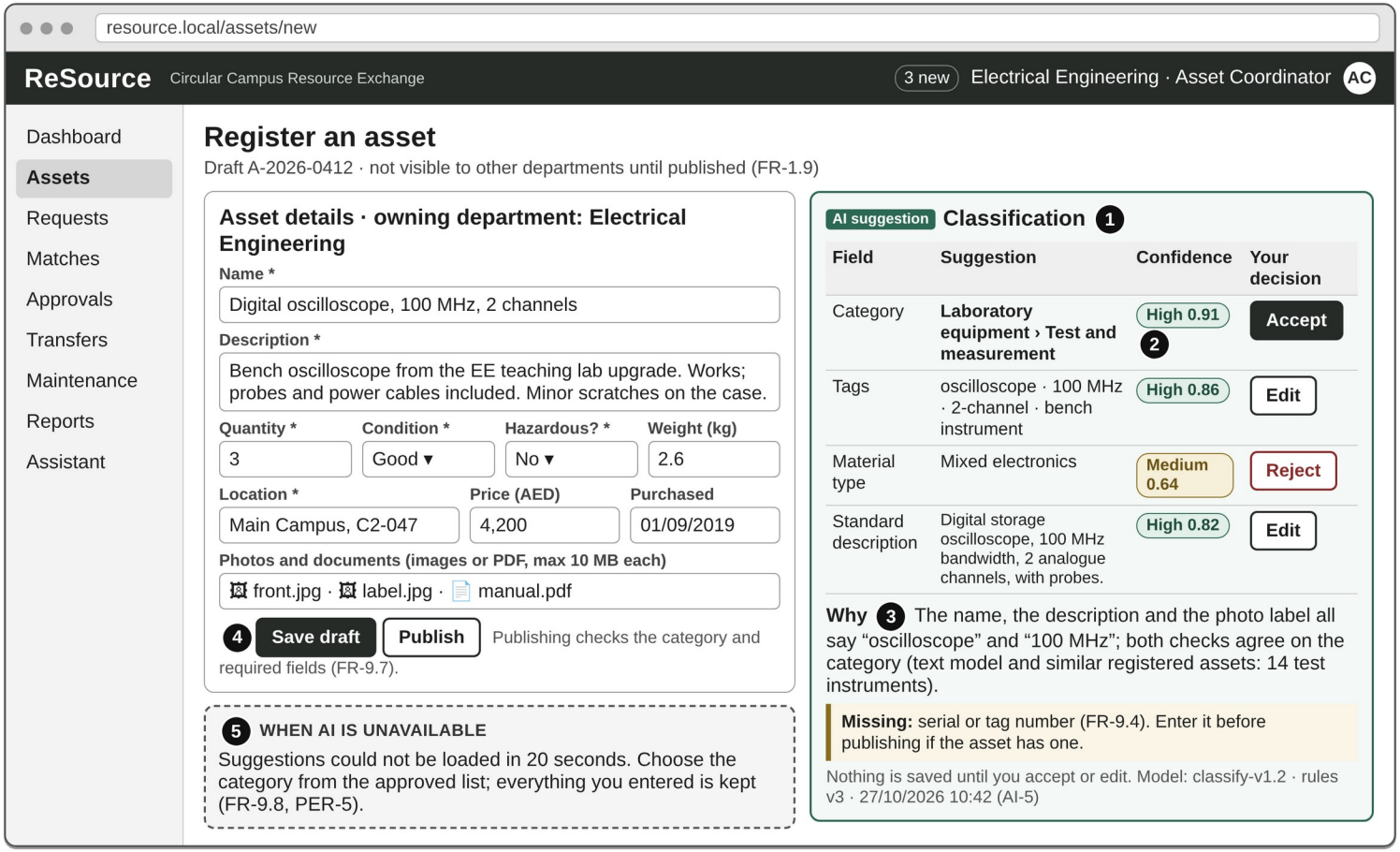


Figure 7-1: Asset Registration screen

7.2 Request Submission

A Physics Department Representative describes the need. While typing, the page shows listings that may already meet it, with a fit score, a confidence level and the reason (FR-8.1). The representative can request a listing directly (UC-3) or submit an open need for the Procurement Team (UC-9).

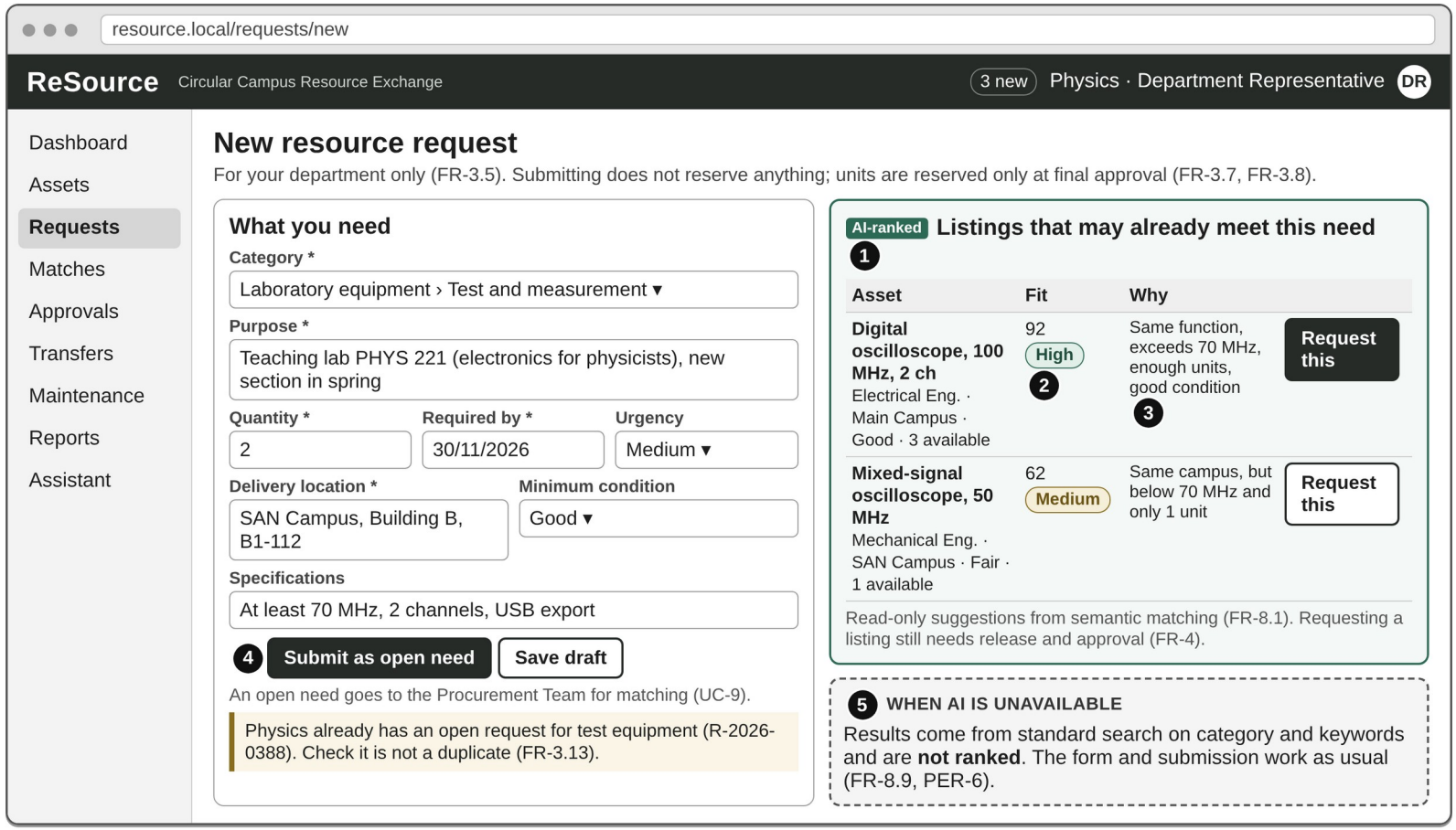


Figure 7-2: Request Submission screen

7.4 Transfer Approval

The Administrator approves the request released by the owning Asset Coordinator. The system shows that the quantity has just been rechecked and that the approver did not submit or release the request (FR-4.4, FR-4.5). The AI score is shown as background only; the decision is the approver's.

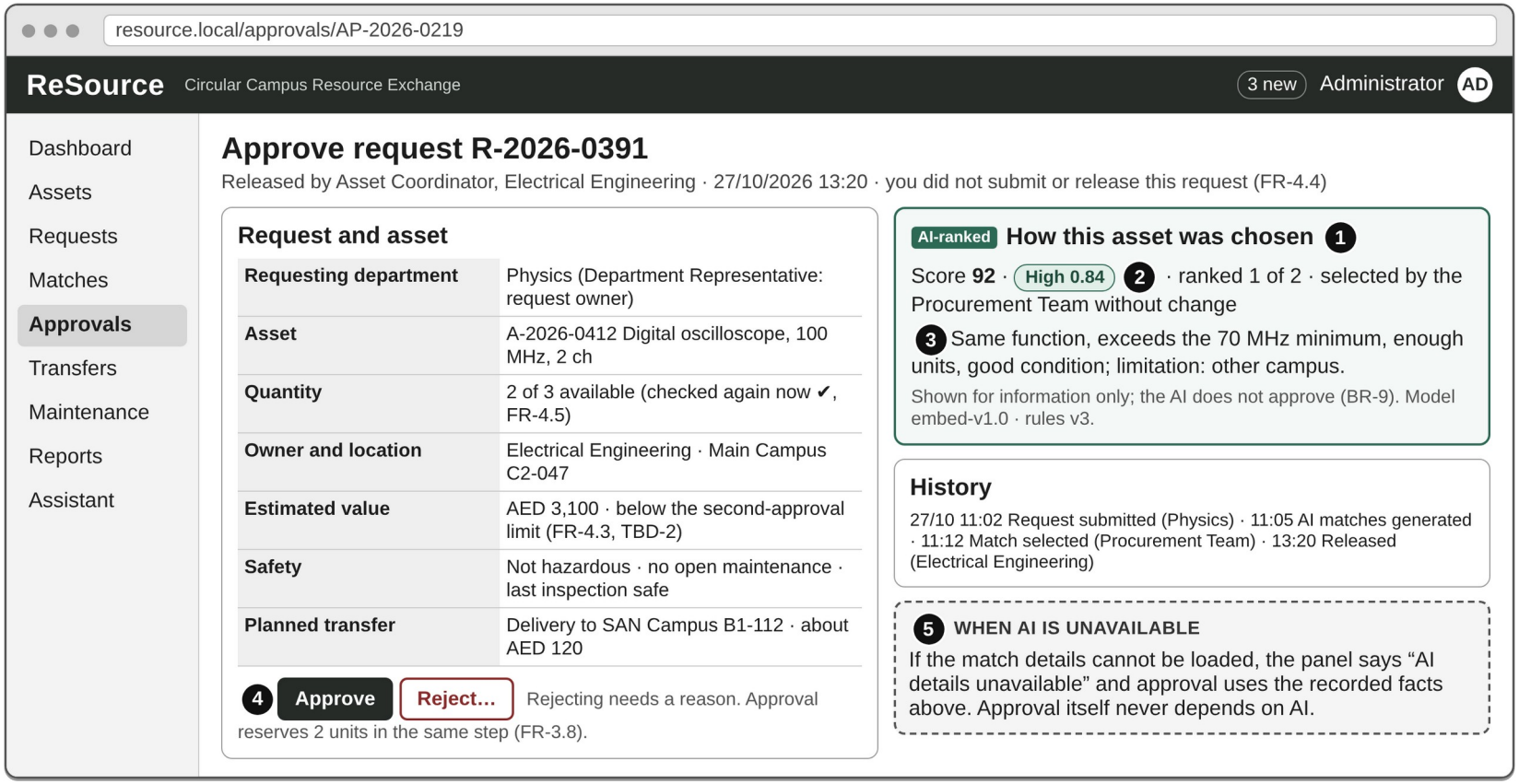


Figure 7-4: Transfer Approval screen

9. Class Diagrams

The class model is drawn in six diagrams: the core domain (Figure 9-1); approvals and transfers (Figure 9-2); maintenance and end of life (Figure 9-3); the AI services (Figure 9-4) and the AI records they keep (Figure 9-5); and notifications, audit and sustainability (Figure 9-6). Every association has a name and a multiplicity at both ends; the name is the relationship name used in the ERD (Section 15) and is read in the direction of the small arrow. Every class that a person creates, decides on or receives is linked to User by a named association. Attributes are private (-) and operations public (+); the operations change state and match the transitions in Section 12. A class drawn in more than one diagram is the same class, shown with only the members needed there, and each class has the name of its table in Section 15 (for example Request and REQUEST).

9.1 Core Domain

Figure 9-1 shows accounts, the catalogue and requests.

- **Aggregation.** A Department employs its Users, but a user record survives a department change and a department cannot be deactivated while it has active users (FR-6.9).
- **Association class.** UserRole records who assigned a role and why (FR-6.8, BR-4). A User gives one or more ConsentRecords, and a withdrawal is a new state of the record, not a deletion (FR-6.2, SEC-8).
- **Composition.** An Asset owns its Location value and its files (up to 5 photos and 3 documents, so 0..8, FR-1.6); a Request owns its RequestSpecification. The parts are deleted only with their owner.
- **Generalization.** ITAsset and FurnitureAsset specialize Asset with their own fields. They are stored in the one ASSET table with an asset_kind column (single-table inheritance), so search and matching treat all assets alike.
- **Navigability.** A Request points to 0..1 Asset: an open need (FR-3.2) has no asset until the Procurement Team selects a match (UC-9).

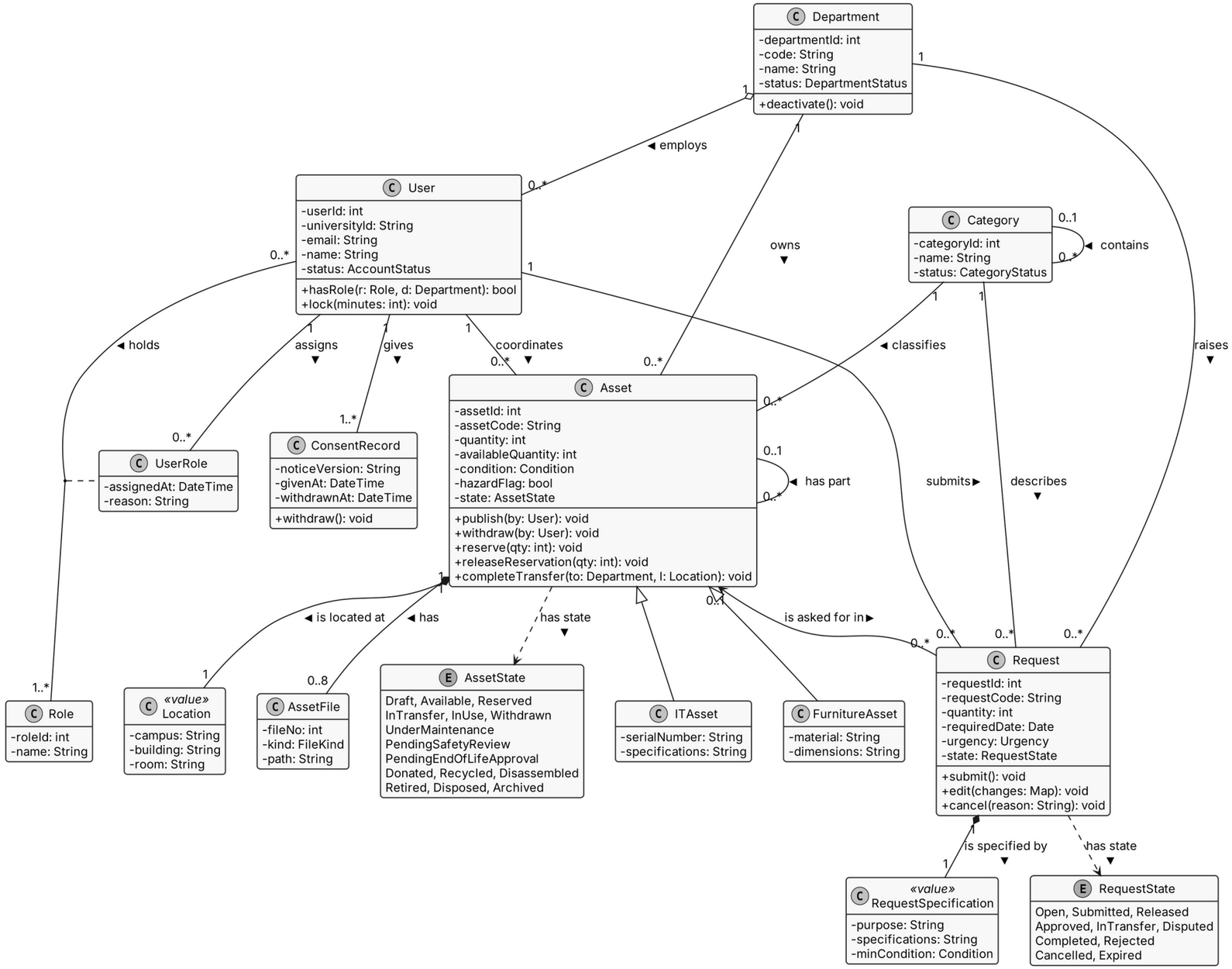


Figure 9-1: Class diagram, core domain

9.2 Approvals, Transfers and Maintenance

A Request has 0 to 3 Approvals: release, approval and, when FR-4.3 applies, the second approval. Each Approval is linked to the User who decides it, and `Approval.isBy(user)` lets the service refuse a person who already filled another stage (FR-4.4). A Reservation exists only after the final approval (BR-5); a Transfer holds up to 5 documents (FR-4.8) and is linked to the User who receives it.

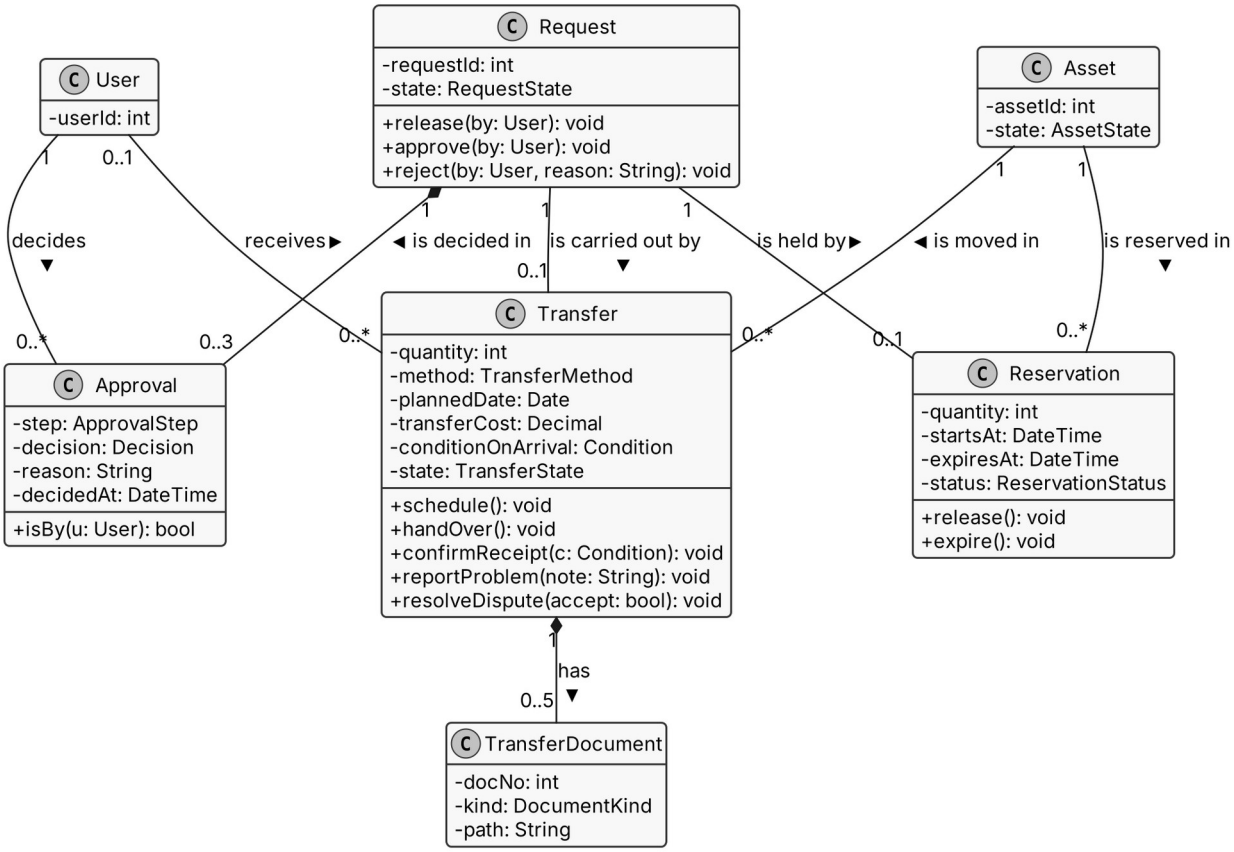


Figure 9-2: Class diagram, approvals and transfers

Inspections, MaintenanceRecords and LifecycleEvents are composed into their owners and have no update operation, because recorded results and life-cycle events cannot be edited (FR-5.18, SEC-12); a mistake is fixed by a new event that corrects the old one. An EndOfLifeProposal is linked to three different Users (proposes, reviews safety of, approves), which is how BR-8 is checked, and to a Vendor only when a vendor handles it; `Vendor.isApproved(date)` blocks expired vendors (FR-5.25).

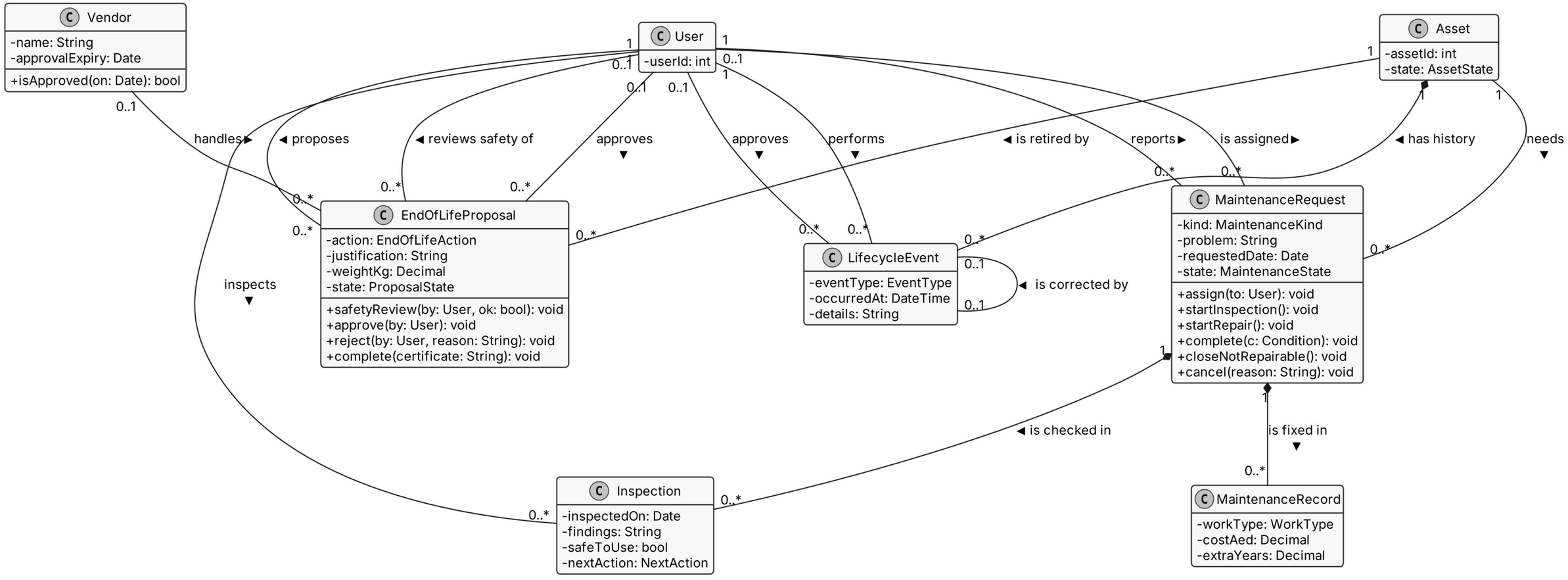


Figure 9-3: Class diagram, maintenance and end of life

10. Sequence Diagrams

10.1 Release and Approve a Request (UC-4)

The release and the approval are separate decisions by different people (FR-4.4, BR-4), and each may end the request. A decline or rejection records the reason, closes the request and notifies the requester; nothing else happens after it. When the value is above the limit or unknown, the Procurement Team gives a second approval (Figure 10-2) before any units are reserved (FR-4.3). The final approval locks the asset row, rechecks the units and reserves them in one transaction, so the last unit can never go to two requests (FR-4.5, QUA-4).

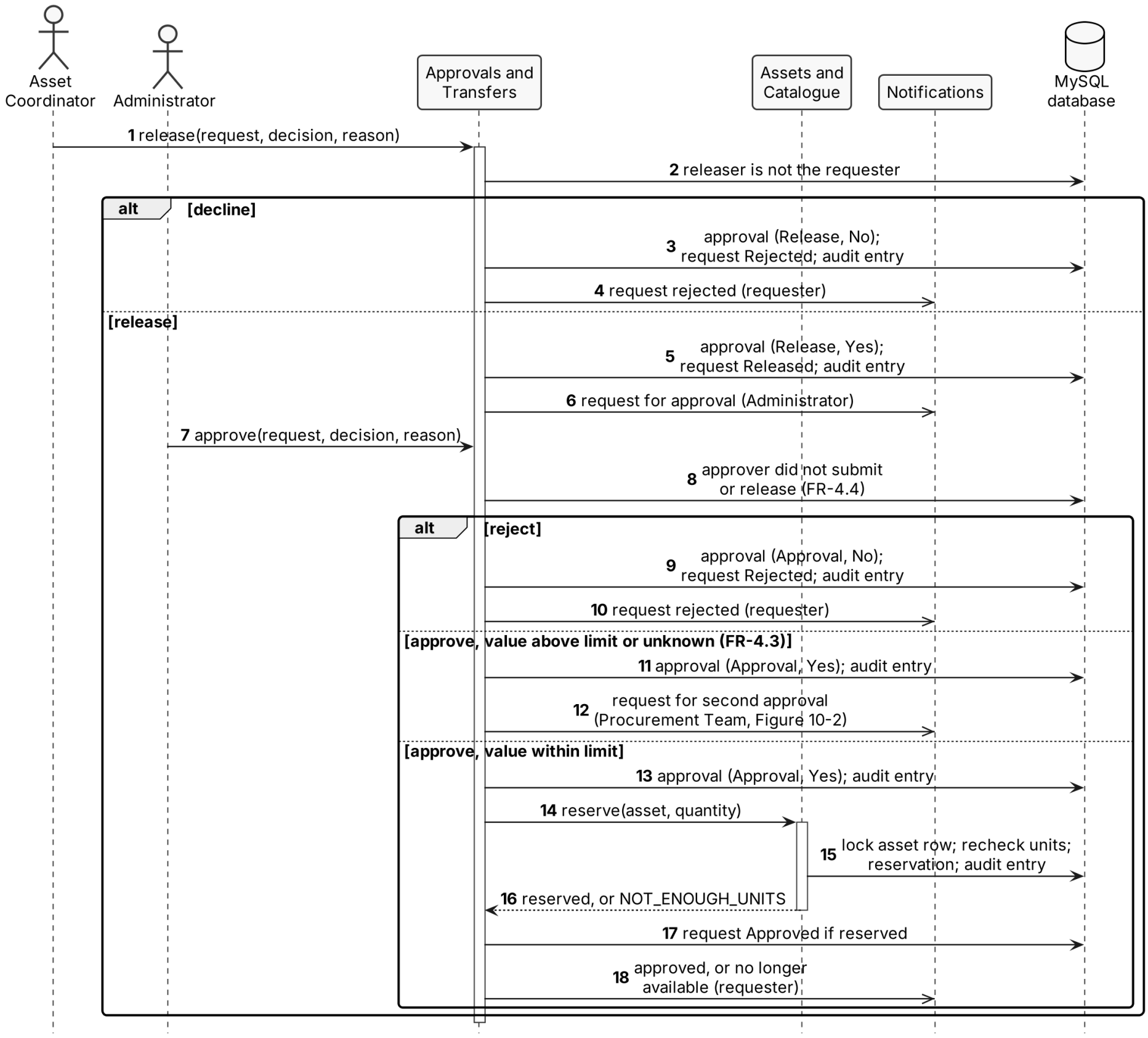


Figure 10-1: Release and approve a request

The second approval is given by a third person, who did not submit, release or approve the request (BR-4). A rejection closes the request; an approval reserves the units exactly as in Figure 10-1.

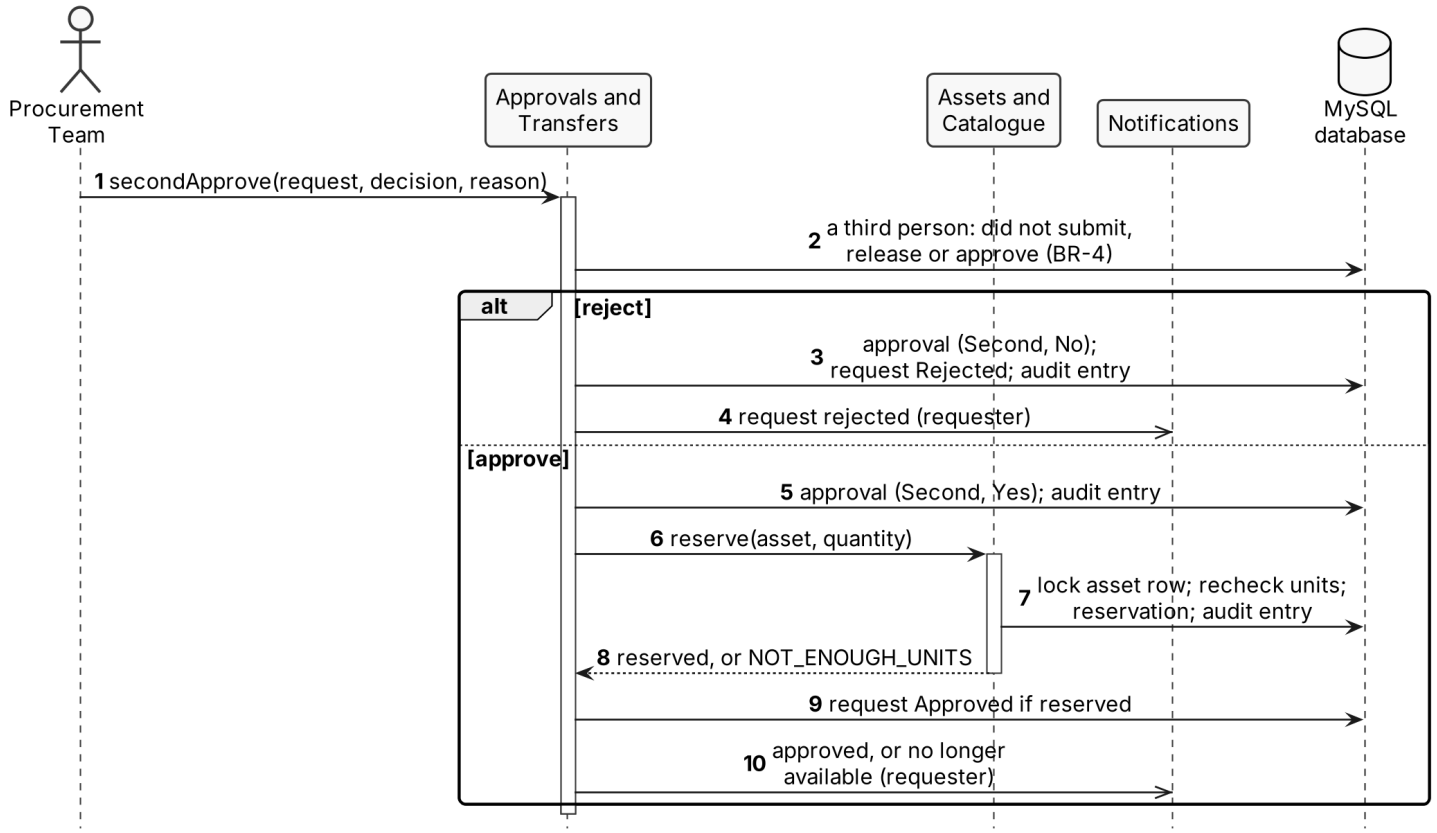


Figure 10-2: Second approval

10.2 Transfer an Asset (UC-5)

Handover is blocked for unsafe or uninspected risky assets (FR-4.16, SAF-9), and handling notes are shown to the people involved (SAF-7). Custody, owning department and location change only when receipt is confirmed or the Administrator accepts a disputed transfer (FR-4.11, FR-4.12); a returned asset becomes available again and the request is cancelled.

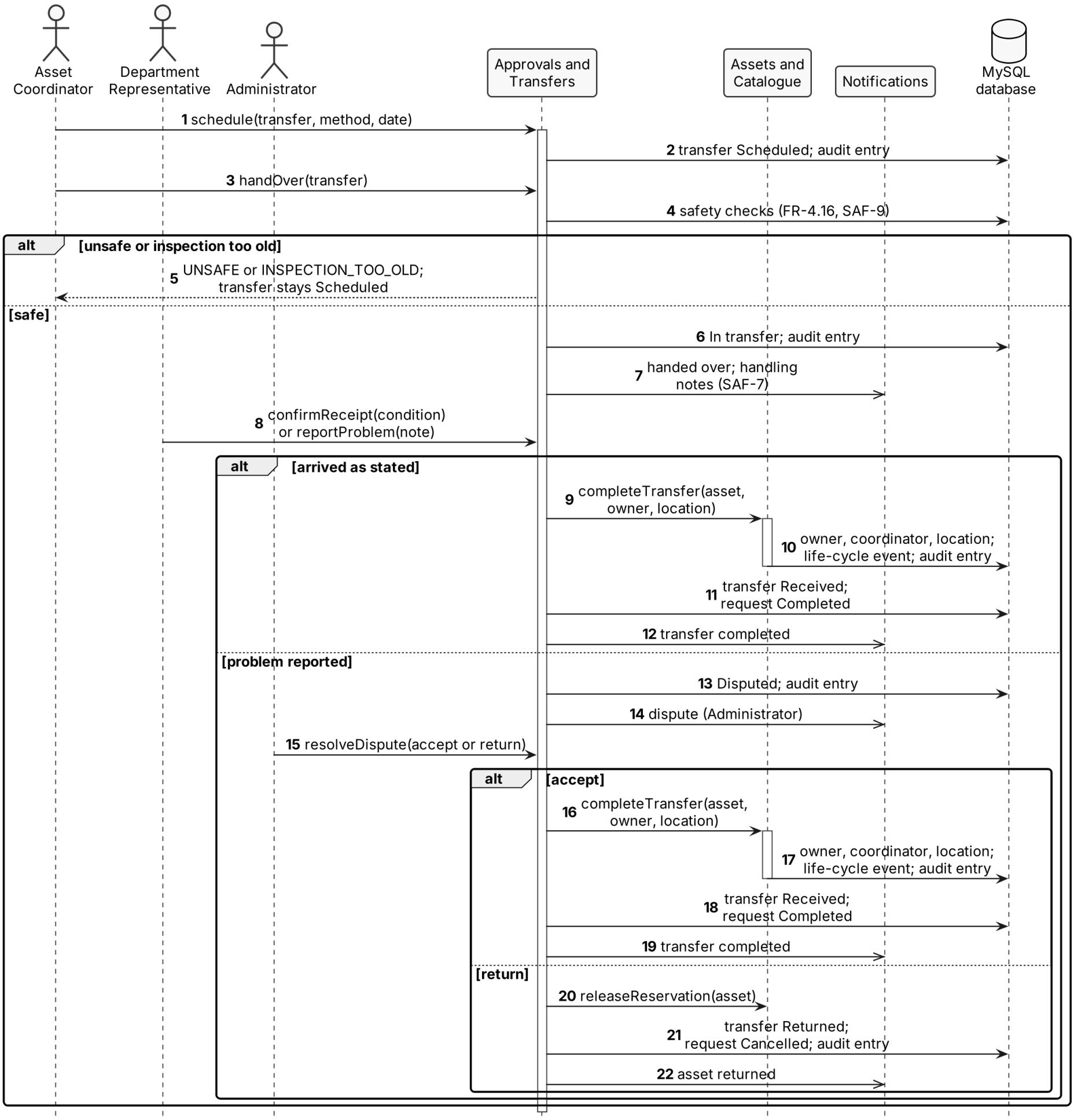


Figure 10-3: Transfer an asset

10.7 Search and Request a Listing (UC-2, UC-3)

Search returns only published assets that the user may see (FR-2.2, FR-2.5). A listing that fits is requested directly; every check runs before anything is saved, and a failed check returns its error code. When nothing fits, the user describes the need and it is saved as an open need for matching (FR-3.2).

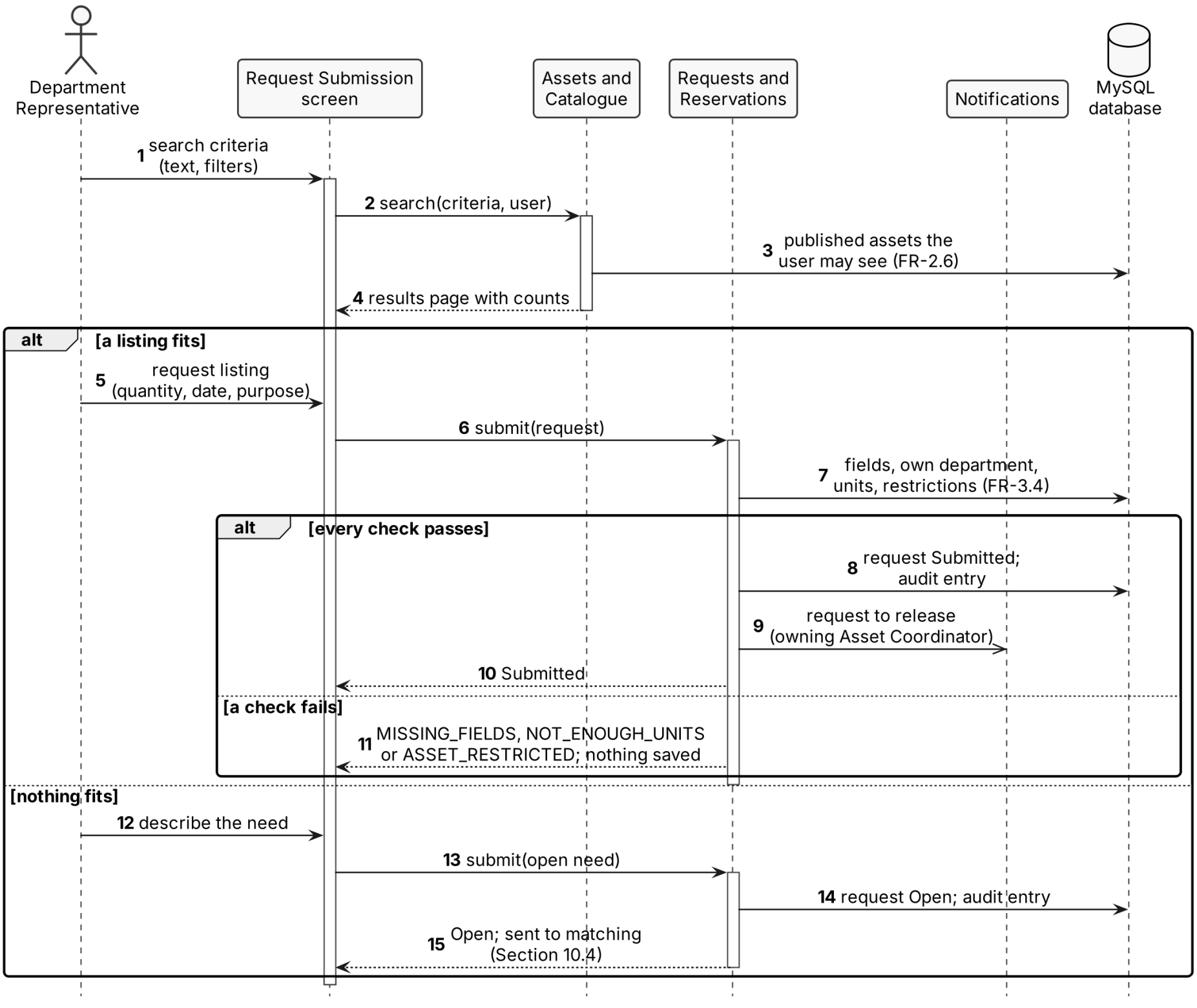


Figure 10-8: Search and request a listing

11. Activity Diagrams

Each swimlane is a role or ReSource itself; a diamond is a decision by the person in that lane, and every "no" branch that ends the flow records a reason.

11.1 From Listing to Approved Request (UC-1 to UC-4, UC-9)

A listing can be requested directly; an open need goes to the Procurement Team, who choose an AI-ranked match or reject all. Both paths then need release and approval, and units are reserved only after the final recheck (FR-4.5).

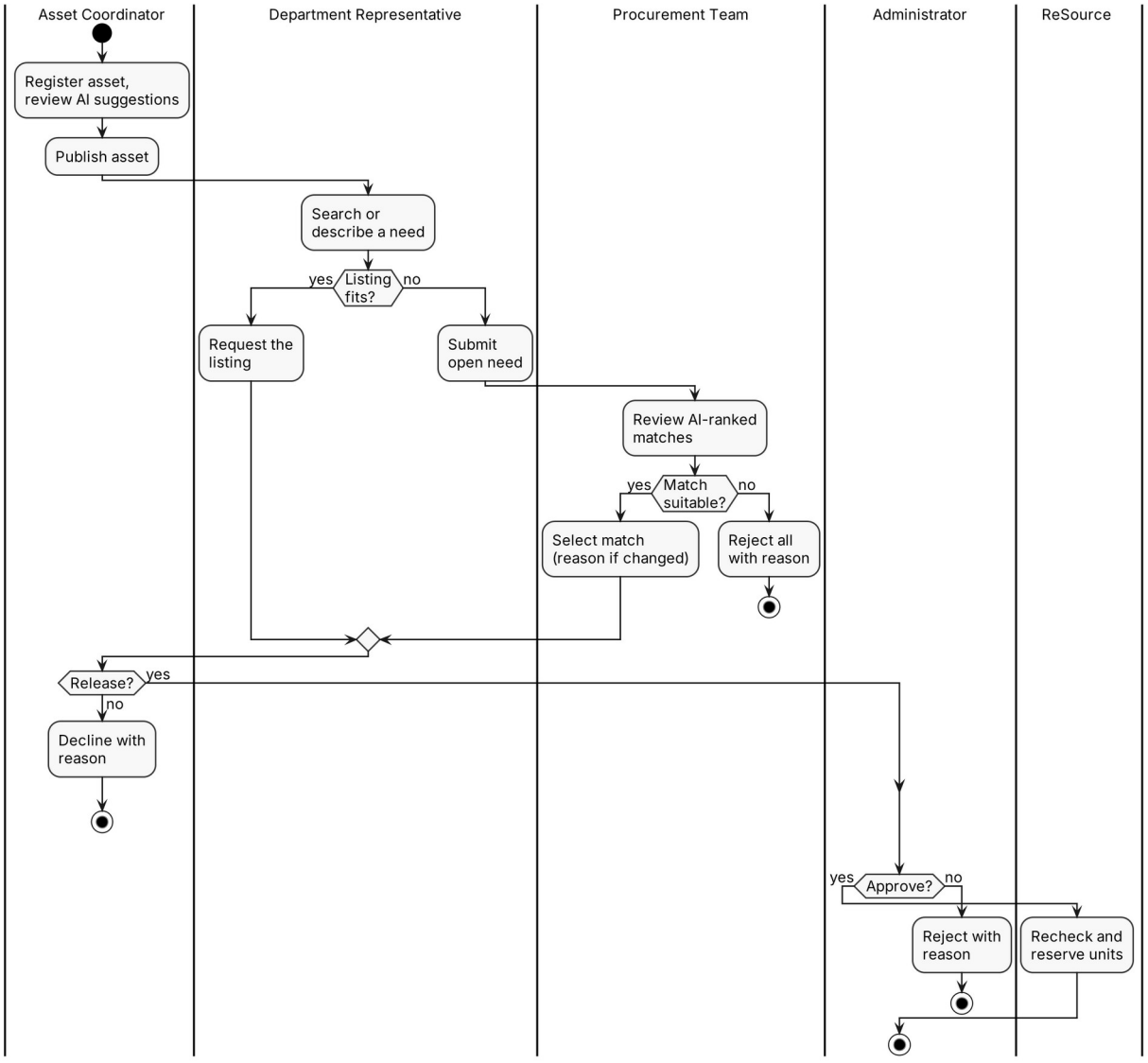


Figure 11-1: Listing to approved request

11.2 Transfer and Receipt (UC-5)

After approval, the owning Asset Coordinator schedules and hands over the asset. A problem on arrival goes to the Administrator, who accepts the transfer or returns the asset (FR-4.12). Custody, owner and location change only on a confirmed receipt or an accepted dispute, and only then do the indicators count the reuse (FR-4.11, FR-13.3).

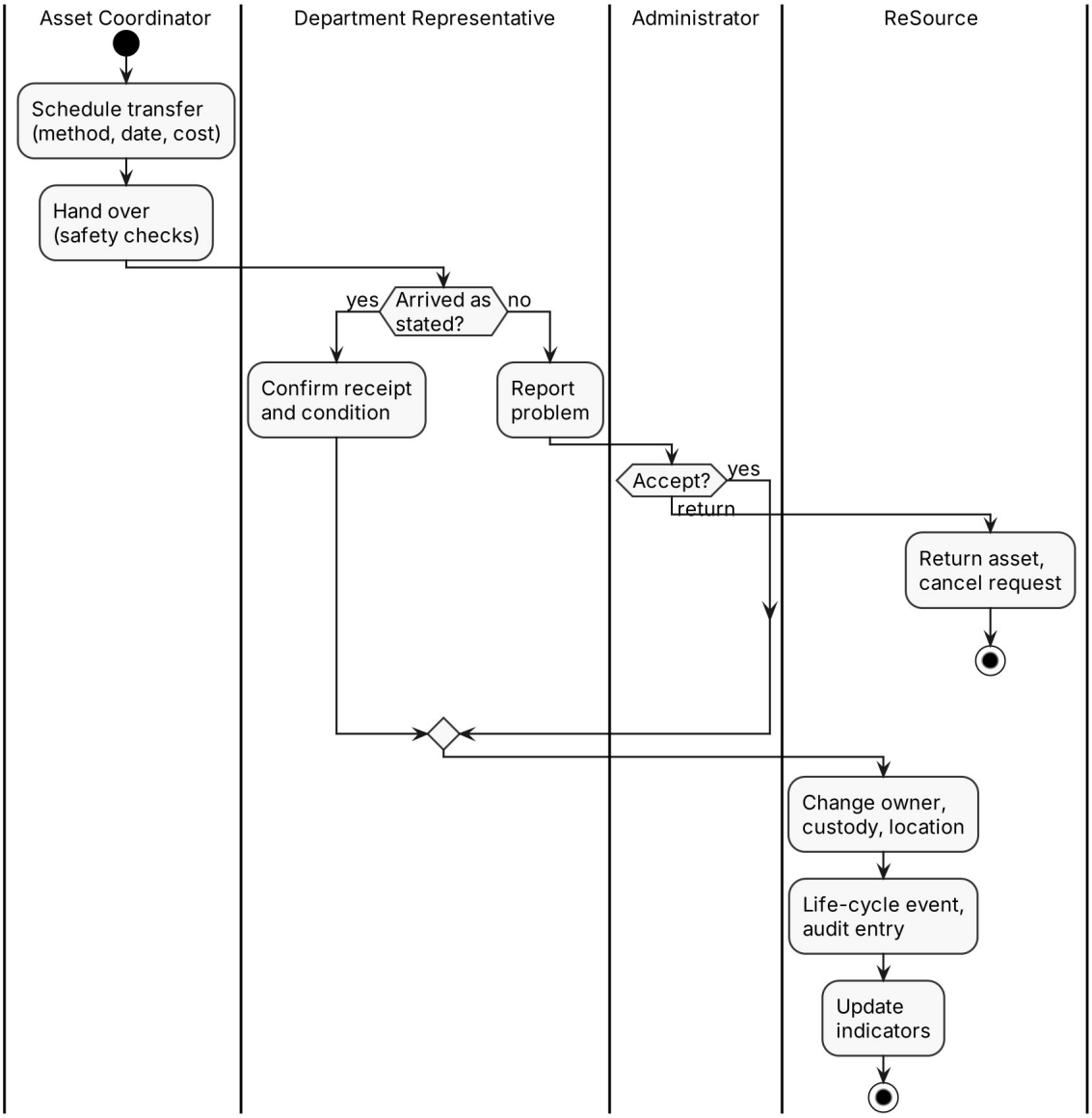


Figure 11-2: Transfer and receipt

12. State Diagrams

The state diagrams use the state lists of Phase 3, Appendix B. Each transition is labeled "event [guard] / action (who)": the guard in brackets must be true, the action after the slash is done by the system, and the role in brackets causes the event, using the abbreviations of Table I-1 ("receiver" is the receiving Department Representative or Asset Coordinator). Approval states are drawn as the first part of the transfer diagram (Figure 12-4), because an approval exists only within one request.

12.1 Asset

Availability is never set by hand; it follows from these states (FR-1.4). Category, quantity and condition cannot change while the asset is Reserved or In transfer (FR-1.10).

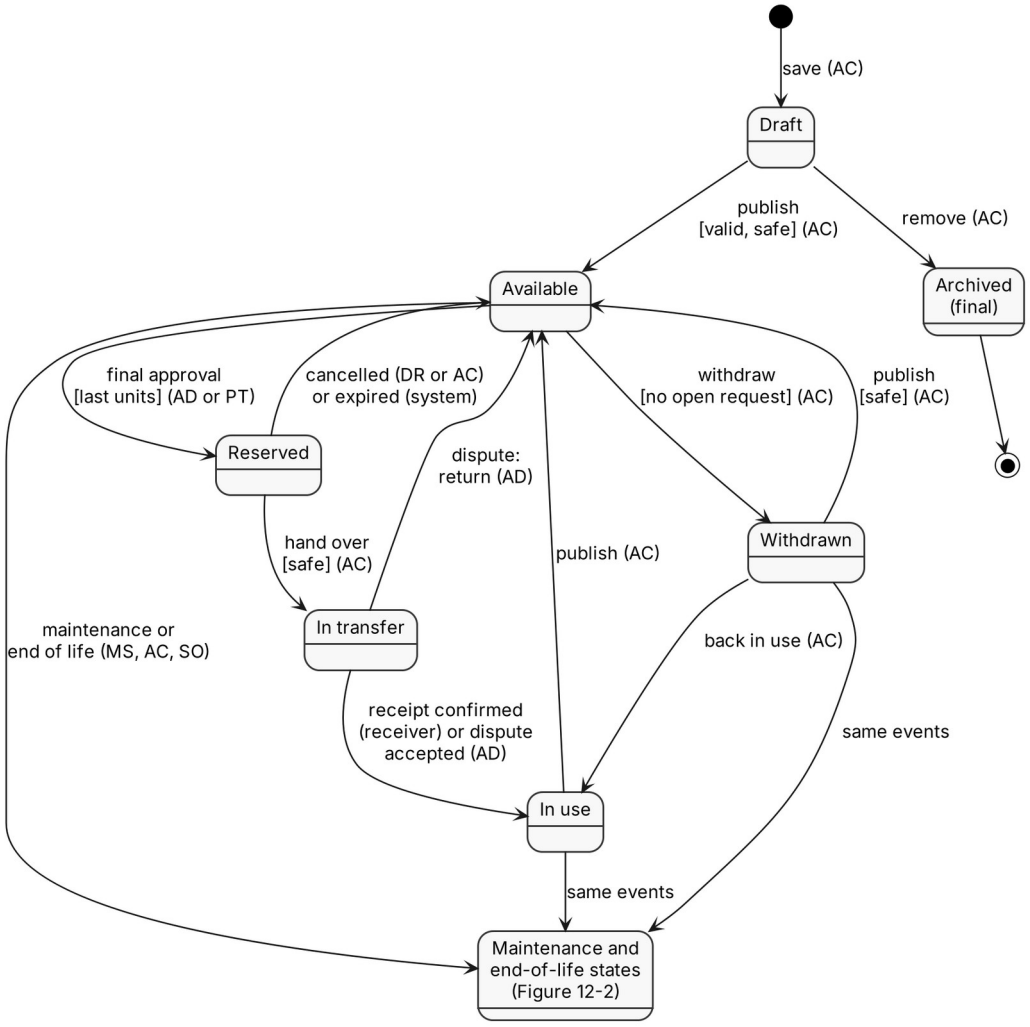


Figure 12-1: Asset states

12.3 Request

A request for a listing starts as Submitted; an open need starts as Open and becomes Submitted when the Procurement Team selects a match. Units are reserved on approval and released on cancellation or expiry (BR-5, FR-3.9).

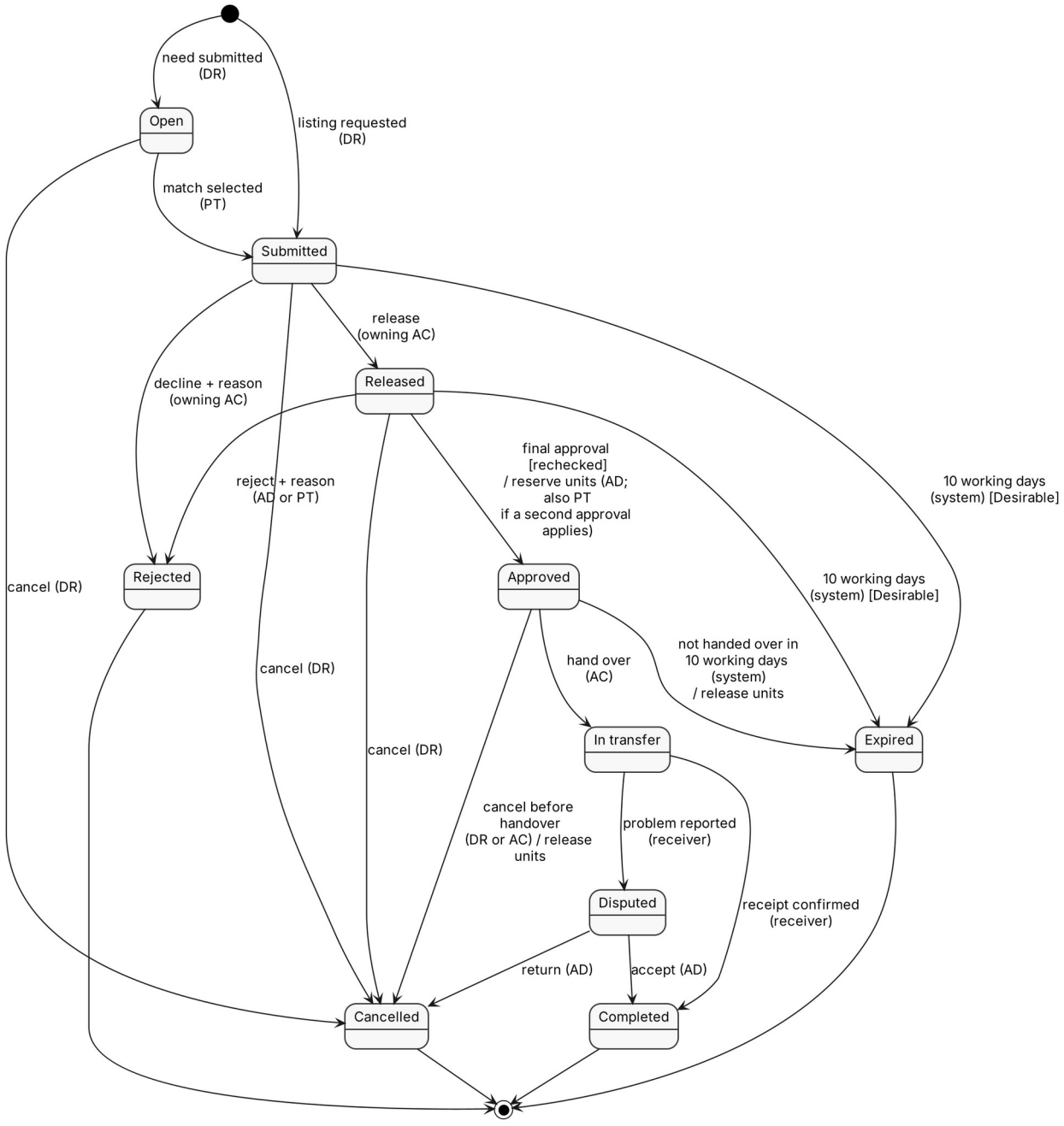


Figure 12-3: Request states

12.4 Approval and Transfer

The approval stages are Awaiting release, Awaiting approval and, for a value above the limit or unknown, Awaiting second approval (FR-4.1 to FR-4.3). Units are reserved when the last required stage approves (BR-5).

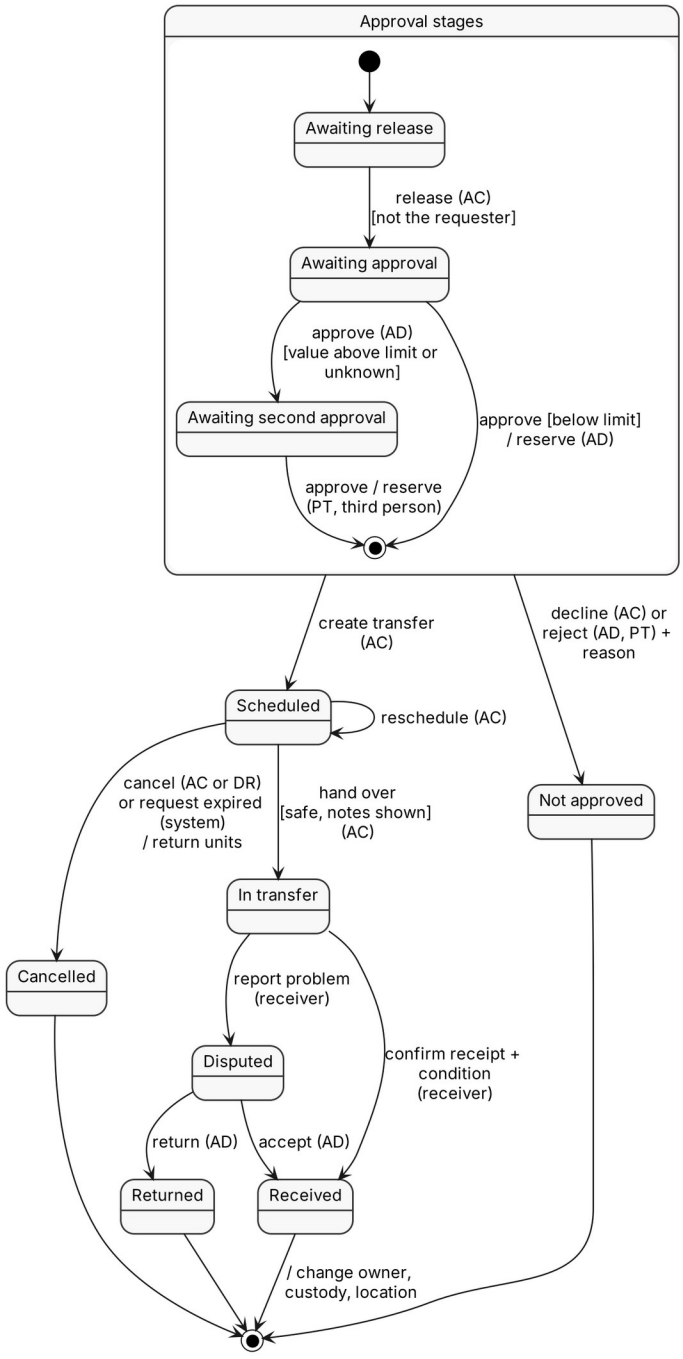


Figure 12-4: Approval and transfer states

15. Entity Relationship Diagram

The ERD has 32 tables and 61 foreign keys, drawn in five parts: accounts and catalogue (Figure 15-1); requests, approvals and transfers (Figure 15-2); maintenance, end of life and life-cycle events (Figure 15-3); AI records (Figure 15-4); and sustainability, notifications and audit (Figure 15-5). Every column is shown with its MySQL type and its keys (PK, FK, UQ). Every foreign key is drawn as its own line, labeled with the relationship name and the foreign key column, with the cardinality at both ends. A grey entity is a table drawn in full in the figure named under it; it is repeated so that each foreign key line can be drawn. Table 15-1 lists the same columns.

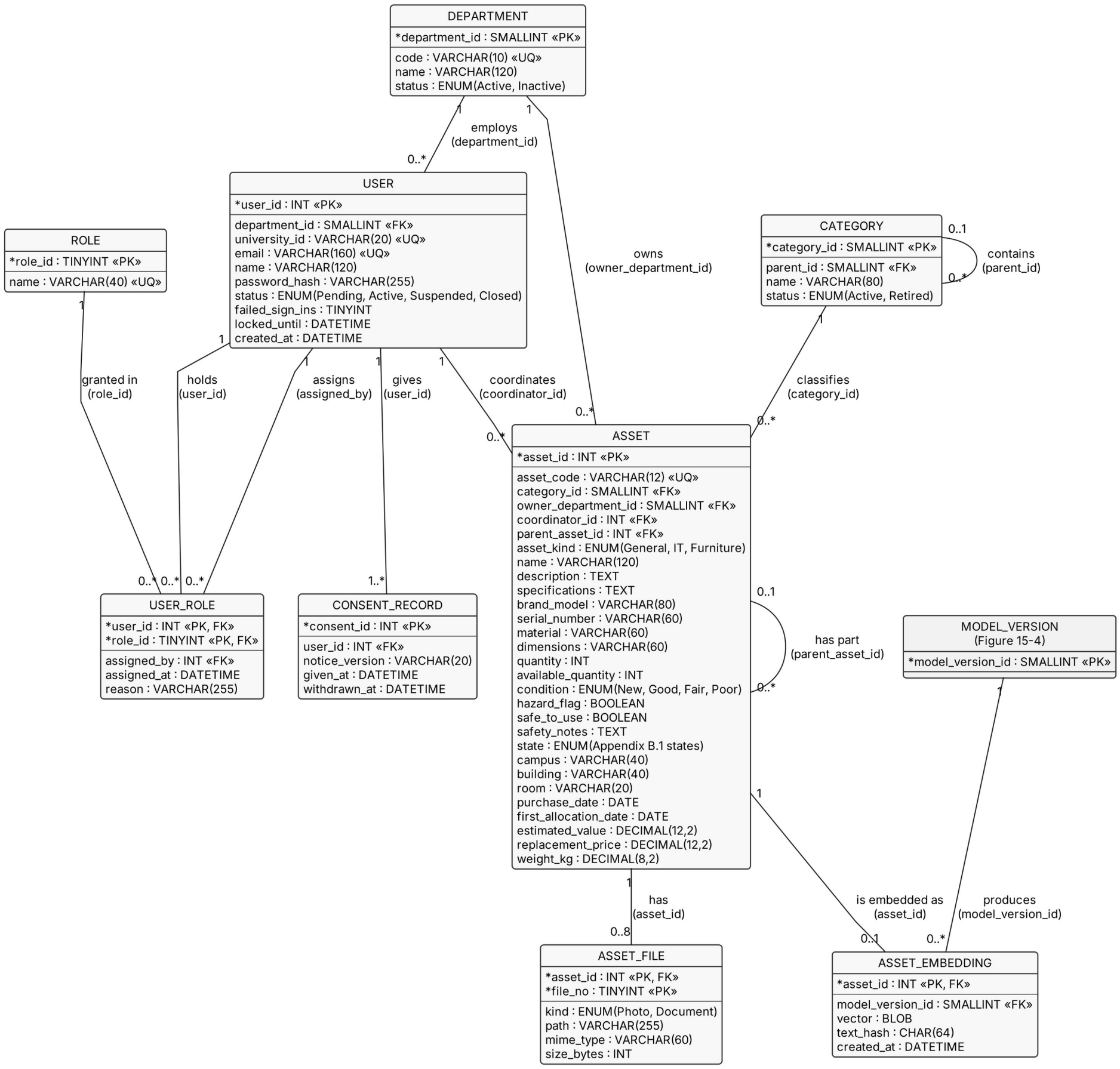


Figure 15-1: ERD part 1, accounts and catalogue

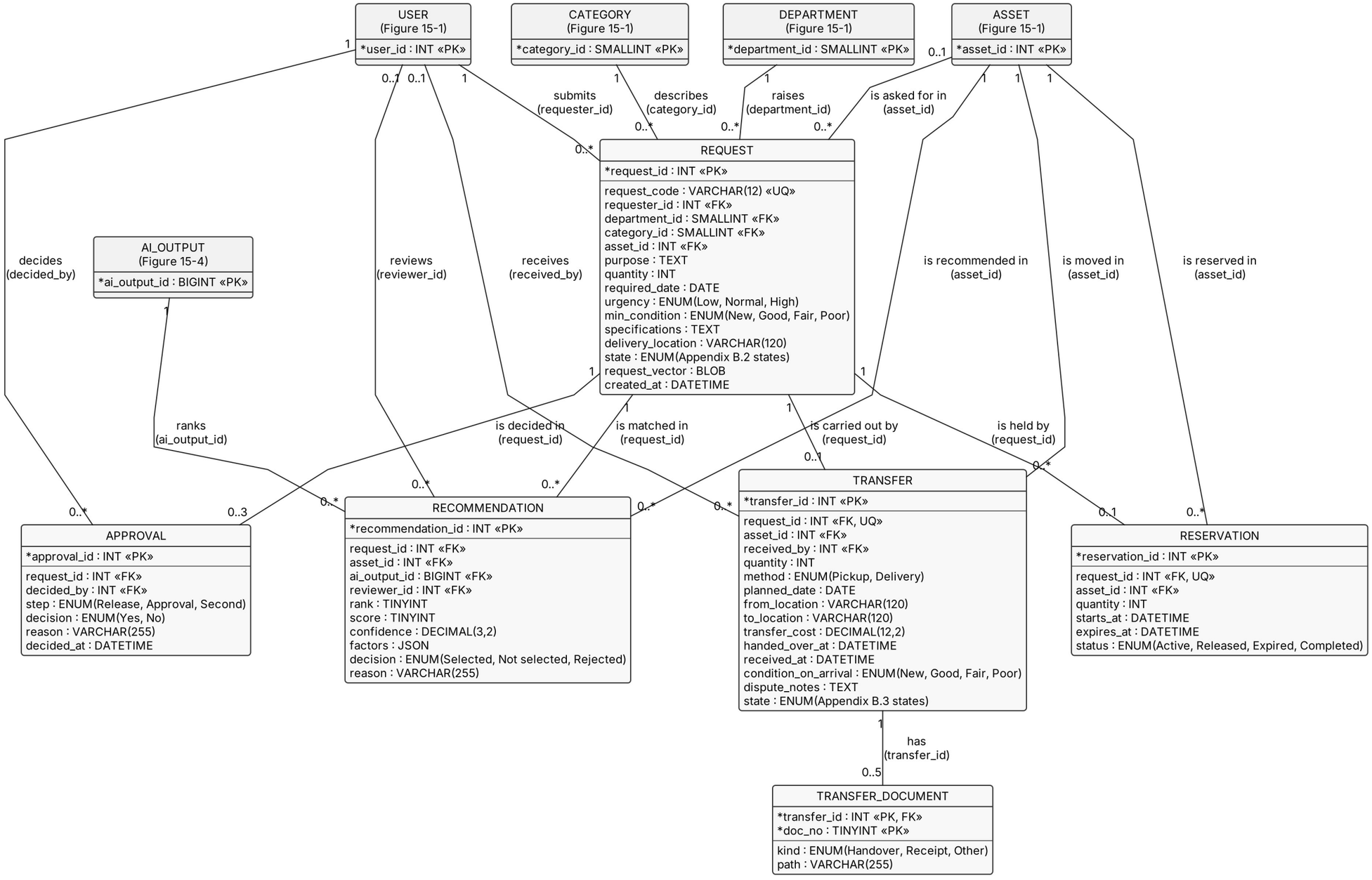


Figure 15-2: ERD part 2, requests, approvals and transfers

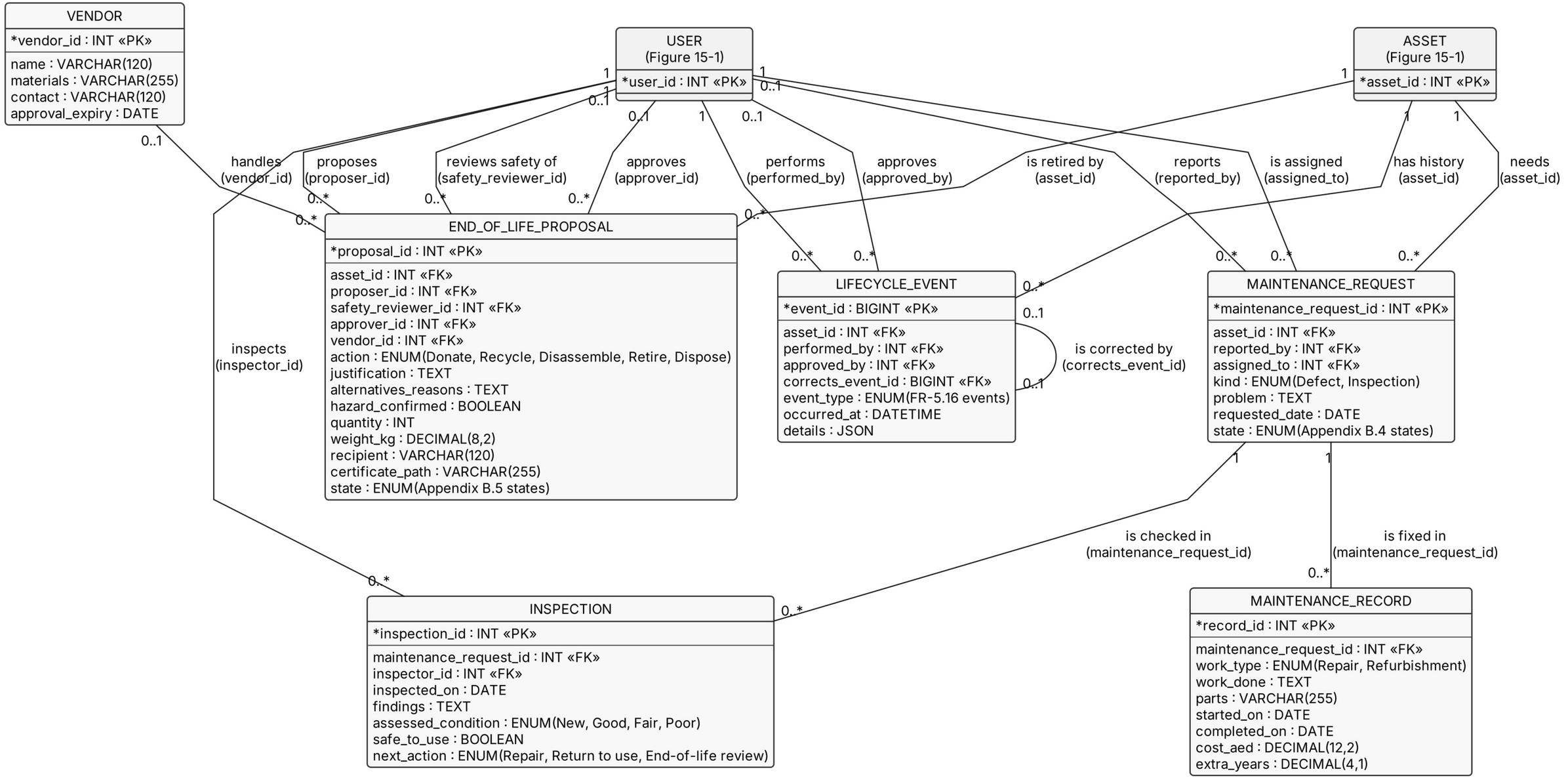


Figure 15-3: ERD part 3, maintenance, end of life and life-cycle events

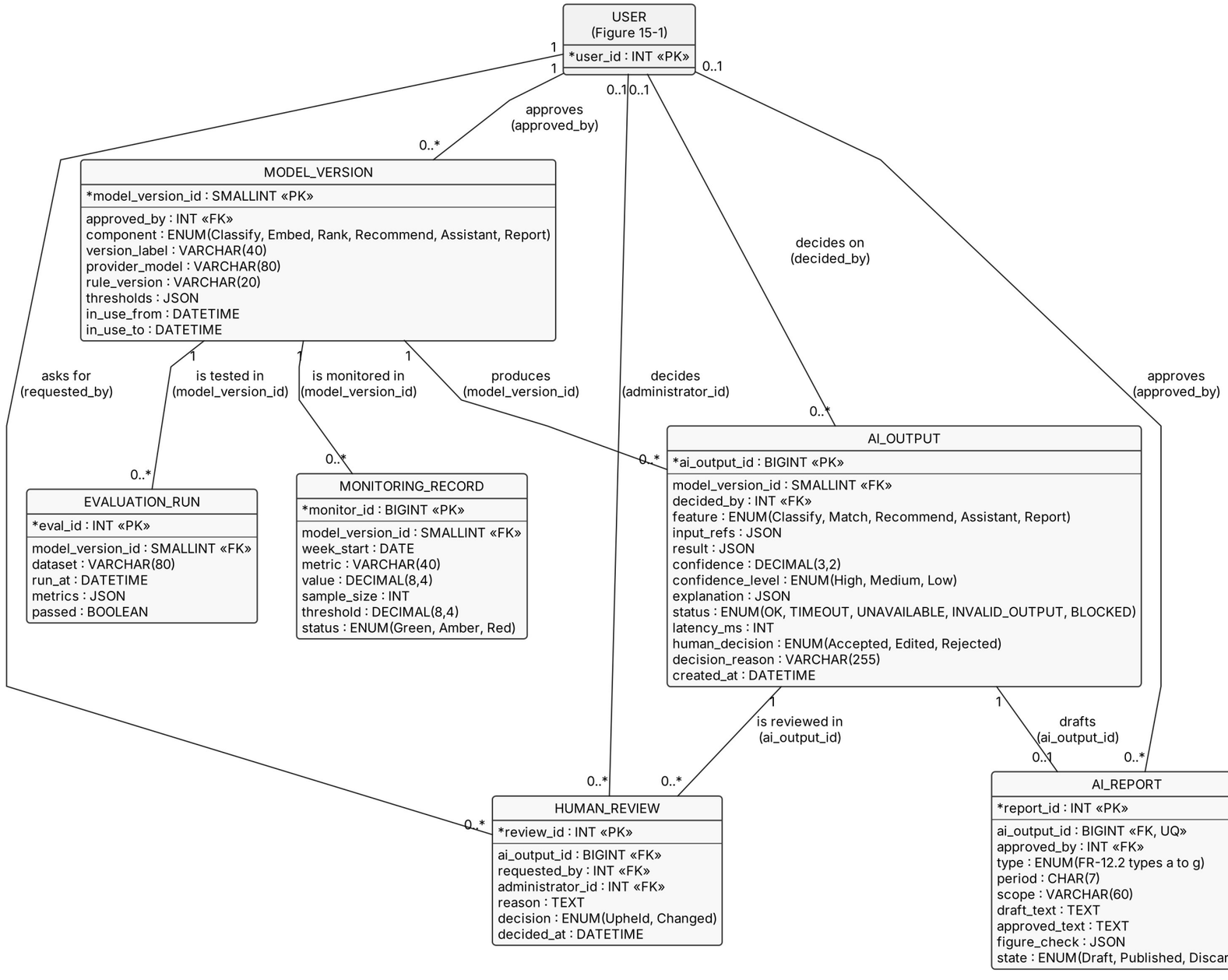


Figure 15-4: ERD part 4, AI records

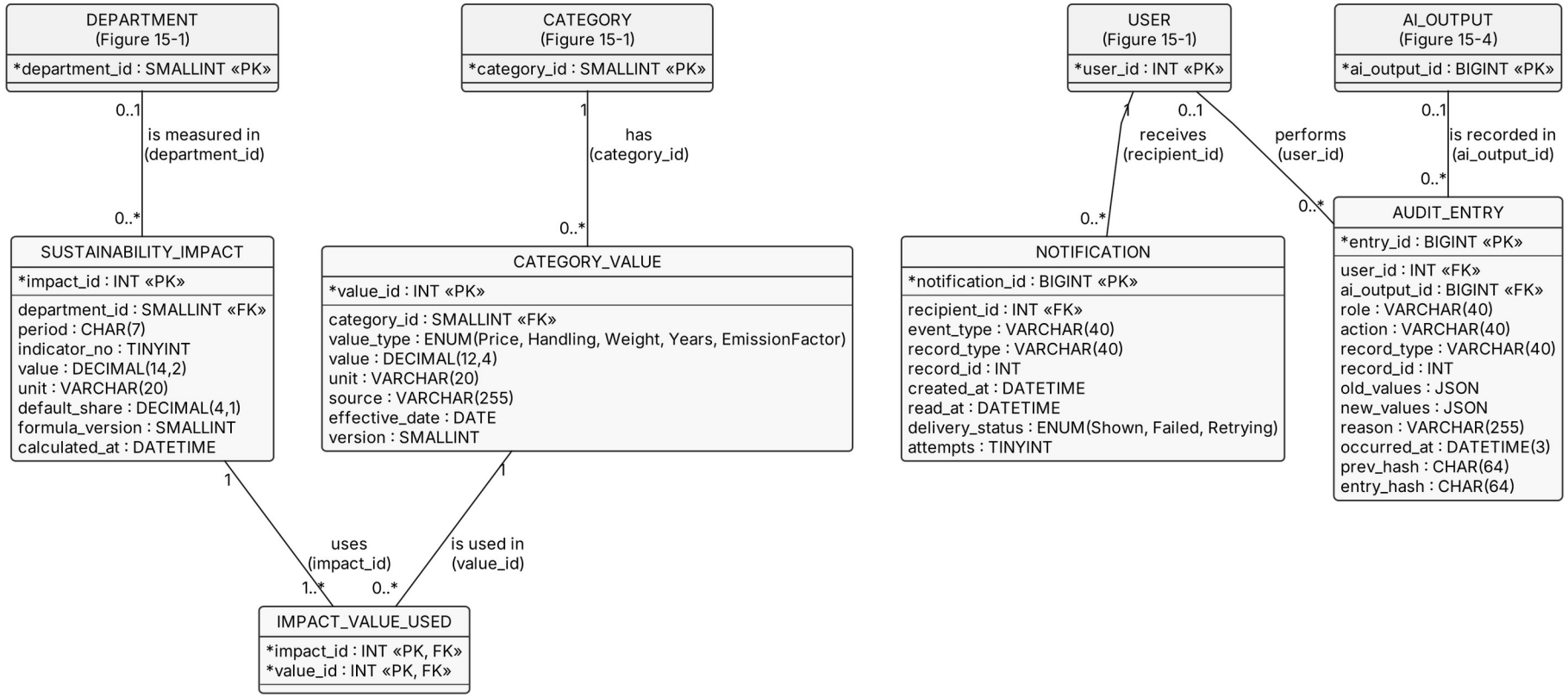


Figure 15-5: ERD part 5, sustainability, notifications and audit

15.1 Notation and Data Dictionary

- **Weak entities.** ASSET_FILE and TRANSFER_DOCUMENT use their owner key plus a partial key (file_no, doc_no).
- **Composite and derived attributes.** Location is stored as campus, building and room. Age is derived from purchase_date and today, so it is calculated, not stored (FR-5.19).
- **Many-to-many links.** USER_ROLE links users and roles; RECOMMENDATION links requests and assets and stores the score, explanation and model version, following [3]; IMPACT_VALUE_USED links each indicator result to the category value versions it used (FR-13.5).
- **Types.** ENUM columns list their values, or name the Phase 3 list they use (Appendix B states, FR-5.16 events, FR-12.2 report types). Money columns are DECIMAL(12,2) in AED and all dates are stored in UAE time.

Table 15-1: Tables and columns

Table	Holds	Keys (PK; FK)	Other columns	Requirements
DEPARTMENT	Campus departments	department_id	code, name, status	FR-6.9
USER	Accounts	user_id; department_id	university_id, email, name, password_hash, status, failed_sign_ins, locked_until, created_at	FR-6.1, SEC-6, SEC-7
ROLE	The nine roles	role_id	name	FR-6.8
USER_ROLE	Role assignments	user_id + role_id; assigned_by	assigned_at, reason	FR-6.8, BR-4
CONSENT_RECORD	Consent given and withdrawn	consent_id; user_id	notice_version, given_at, withdrawn_at	FR-6.2, SEC-8
CATEGORY	Approved category list	category_id; parent_id	name, status	FR-9.2, QUA-7
ASSET	Registered assets	asset_id; category_id, owner_department_id, coordinator_id, parent_asset_id	asset_code, asset_kind, name, description, specifications, brand_model, serial_number, material, dimensions, quantity, available_quantity, condition, hazard_flag, safe_to_use, safety_notes, state, campus, building, room, purchase_date, first_allocation_date, estimated_value, replacement_price, weight_kg	FR-1.2 to FR-1.10, FR-4.15
ASSET_FILE	Photos and documents (weak entity)	asset_id + file_no	kind, path, mime_type, size_bytes	FR-1.6, SEC-13
ASSET_EMBEDDING	Text vector of an asset	asset_id; model_version_id	vector, text_hash, created_at	FR-8.1
REQUEST	Requests and open needs	request_id; requester_id, department_id, category_id, asset_id	request_code, purpose, quantity, required_date, urgency, min_condition, specifications, delivery_location, state, request_vector, created_at	FR-3.1 to FR-3.15
RECOMMENDATION	One ranked match (links requests and assets)	recommendation_id; request_id, asset_id, ai_output_id, reviewer_id	rank, score, confidence, factors, decision, reason	FR-8.4, FR-8.7, QUA-12
APPROVAL	Release, approval and second approval	approval_id; request_id, decided_by	step, decision, reason, decided_at	FR-4.1 to FR-4.4
RESERVATION	Reserved units	reservation_id; request_id, asset_id	quantity, starts_at, expires_at, status	FR-3.8, FR-4.5
TRANSFER	Movement of an asset	transfer_id; request_id, asset_id, received_by	quantity, method, planned_date, from_location, to_location, transfer_cost, handed_over_at, received_at, condition_on_arrival, dispute_notes, state	FR-4.7 to FR-4.12
TRANSFER_DOCUMENT	Handover files (weak entity)	transfer_id + doc_no	kind, path	FR-4.8
MAINTENANCE_REQUEST	Defects and inspection requests	maintenance_request_id; asset_id, reported_by, assigned_to	kind, problem, requested_date, state	FR-5.1 to FR-5.7
INSPECTION	Inspection results	inspection_id; maintenance_request_id, inspector_id	inspected_on, findings, assessed_condition, safe_to_use, next_action	FR-5.5, SAF-5
MAINTENANCE_RECORD	Repair and refurbishment work	record_id; maintenance_request_id	work_type, work_done, parts, started_on, completed_on, cost_aed, extra_years	FR-5.6, FR-5.10
END_OF_LIFE_PROPOSAL	End-of-life proposal and completion	proposal_id; asset_id, proposer_id, safety_reviewer_id, approver_id, vendor_id	action, justification, alternatives_reasons, hazard_confirmed, quantity, weight_kg, recipient, certificate_path, state	FR-5.20 to FR-5.32, SAF-4
VENDOR	Approved recycling and disposal vendors	vendor_id	name, materials, contact, approval_expiry	FR-5.25
LIFECYCLE_EVENT	Life-cycle history (no updates)	event_id; asset_id, performed_by, approved_by, corrects_event_id	event_type, occurred_at, details	FR-5.16 to FR-5.18

Table	Holds	Keys (PK; FK)	Other columns	Requirements
CATEGORY_VALUE	Versioned prices, costs, weights, years and emission factors	value_id; category_id	value_type, value, unit, source, effective_date, version	FR-13.4, FR-13.5
SUSTAINABILITY_IMPACT	Calculated indicator values	impact_id; department_id	period, indicator_no, value, unit, default_share, formula_version, calculated_at	FR-13.1, FR-13.7
IMPACT_VALUE_USED	Category value versions used in each result	impact_id + value_id	-	FR-13.5
MODEL_VERSION	AI model, rule and threshold versions	model_version_id; approved_by	component, version_label, provider_model, rule_version, thresholds, in_use_from, in_use_to	AI-5, AI-10
AI_OUTPUT	Every AI result and its human decision	ai_output_id; model_version_id, decided_by	feature, input_refs, result, confidence, confidence_level, explanation, status, latency_ms, human_decision, decision_reason, created_at	FR-7.A4, AI-5
HUMAN_REVIEW	Requests for human review of an AI result	review_id; ai_output_id, requested_by, administrator_id	reason, decision, decided_at	SEC-10
AI_REPORT	AI report drafts and approved versions	report_id; ai_output_id, approved_by	type, period, scope, draft_text, approved_text, figure_check, state	FR-12.6 to FR-12.10
EVALUATION_RUN	Test-set evaluations of a model version	eval_id; model_version_id	dataset, run_at, metrics, passed	QUA-9, QUA-10
MONITORING_RECORD	Weekly monitoring metrics	monitor_id; model_version_id	week_start, metric, value, sample_size, threshold, status	QUA-12
NOTIFICATION	In-application notifications	notification_id; recipient_id	event_type, record_type, record_id, created_at, read_at, delivery_status, attempts	FR-7.N1 to FR-7.N5
AUDIT_ENTRY	Append-only audit trail	entry_id; user_id, ai_output_id	role, action, record_type, record_id, old_values, new_values, reason, occurred_at, prev_hash, entry_hash	FR-7.A1 to FR-7.A6

Django-Q2 adds its own job tables, and Django adds its session table. The data stores of Section 2.2 map to these tables as shown in Table 2-2.

16. Interface Contracts

Section 16.1 defines the operations that modules offer each other and the web pages; Section 16.2 defines the contracts between the modules and the AI services. All operations follow four common rules:

1. Every operation receives the acting user and checks access first (AccessCheck); a refusal raises `PERMISSION_DENIED` and is audited (FR-6.7, FR-7.A1).
2. Every operation that changes data runs in one database transaction with its audit entry; if the audit entry fails, nothing is saved (FR-7.A6).
3. Notifications are sent after the transaction commits; a notification failure never reverses the change (FR-7.N3).
4. Errors are returned as a code and a message that names the problem (for example `MISSING_FIELDS: location, condition`).

16.1 Module Operations

Table 16-1: Module service operations

Module	Operation	Input	Output	Errors	Requirements
Accounts and Access	apply, activate, refuse, assignRole, suspend, withdrawConsent	Applicant details and consent; user, role, department, reason	Account state	DUPLICATE, NOT_AUTHORIZED, CONSENT_MISSING	FR-6.1 to FR-6.12
Accounts and Access	allowed(user, action, record)	User, action, record	Yes or no	-	FR-6.7, SEC-5
Assets and Catalogue	saveDraft, publish, withdraw	Asset fields, files	asset_id, state	MISSING_FIELDS, INVALID_VALUE, SAFETY_REVIEW_NEEDED, FILE_REJECTED	FR-1, FR-2.1
Assets and Catalogue	search(criteria)	Text, filters, page	Results page with counts	INVALID_CRITERIA	FR-2.2 to FR-2.6
Assets and Catalogue	reserve, releaseReservation, completeTransfer	asset_id, quantity, request_id or transfer_id	Reservation or updated asset	NOT_ENOUGH_UNITS, RESTRICTED	FR-4.5, FR-4.11, FR-4.15
Requests and Reservations	submit, edit, cancel	Request fields	request_id, state	MISSING_FIELDS, NOT_OWN_DEPARTMENT, NOT_ENOUGH_UNITS, ASSET_RESTRICTED, WRONG_STATE	FR-3.1 to FR-3.15
Match Review	matches(request_id)	request_id	Ranked list or "not ranked" list	NO_ELIGIBLE_ASSETS	FR-8.1 to FR-8.9
Match Review	decide(request_id, asset_id, decision, reason)	Choice and reason	Request sent to release	REASON_REQUIRED, NO_LONGER_AVAILABLE	FR-8.7, FR-8.8
Approvals and Transfers	release, approve, secondApprove	request_id, decision, reason	Approval record; reservation on final approval	SAME_PERSON, REASON_REQUIRED, NOT_ENOUGH_UNITS	FR-4.1 to FR-4.6
Approvals and Transfers	schedule, handOver, confirmReceipt, reportProblem, resolveDispute	transfer details, condition, decision	Transfer state	NOT_APPROVED, UNSAFE, INSPECTION_TOO_OLD, NOT_RECEIVER	FR-4.7 to FR-4.16
Maintenance and End-of-Life	report, assign, startInspection, recordInspection, recordWork, complete, closeNotRepairable, cancel	Maintenance fields	Maintenance state	NOT_MAINTENANCE_ROLE, INVALID_DATES, MISSING_FIELDS	FR-5.1 to FR-5.15
Maintenance and End-of-Life	propose, safetyReview, decide, complete	Action, reasons, review, completion record	Proposal state	BLOCKED_BY_OPEN_WORK, SAME_PERSON, VENDOR_EXPIRED, CERTIFICATE_REQUIRED	FR-5.20 to FR-5.34
Reporting and Indicators	standardReport, indicators, requestAIReport, approveReport, discardReport	Report type, period, scope	Report, indicator set or AI report draft	NO_RECORDS, NOT_AUTHORIZED	FR-7.R, FR-12, FR-13
Notifications	notify(event, record, recipients)	Event and record	Notification IDs	Retried on failure	FR-7.N1 to FR-7.N5
Audit	write(user, action, record, before, after, reason, ai_output_id)	Entry fields	entry_id	AUDIT_FAILED (change rolled back)	FR-7.A1 to FR-7.A6

AI Contributions

I used an AI assistant (ChatGPT astra6, OpenAI) to help draw the class diagrams in Sections 9.1 and 9.2, the sequence, activity and state diagrams in Sections 10.1, 10.2, 10.7, 11.1, 11.2, 12.1, 12.3 and 12.4, the ERD (Section 15) and the screen mockups in Sections 7.1, 7.2 and 7.4 from my design, and to check the states, entities and keys against the Phase 3 report. I wrote the section content and reviewed every figure.

References

[1] University, "Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System," COSC 336 Introduction to Software Engineering project description, University, Abu Dhabi, UAE, Fall 2026.

[2] University, "Phase 4: Architecture Design," COSC 336 lab slides, University, Abu Dhabi, UAE, Fall 2026.

[3] University, "Phase 4 & 5: Design Document, Sequence, ER and Class Diagrams," COSC 336 lab slides, University, Abu Dhabi, UAE, Fall 2026.

[4] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, "ReSource Phase III: Requirements Document," COSC 336, University, Abu Dhabi, UAE, Oct. 2026.

[5] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, "ReSource Phase II Feasibility Study," COSC 336, University, Abu Dhabi, UAE, Sep. 2026.

[6] Django Software Foundation, "User authentication in Django," Django documentation. [Online]. Available: https://docs.djangoproject.com/en/stable/topics/auth/

[7] Anthropic, "Models overview," Claude Platform Docs, 2026. [Online]. Available: https://platform.claude.com/docs/en/about-claude/models/overview

[8] Django Q2 project, "Brokers," Django Q2 documentation. [Online]. Available: https://django-q2.readthedocs.io/en/stable/brokers.html

[9] World Wide Web Consortium, "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Dec. 2024. [Online]. Available: https://www.w3.org/TR/WCAG22/

[10] Sentence Transformers, "all-MiniLM-L6-v2," model card, Hugging Face. [Online]. Available: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[11] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), Hong Kong, 2019, pp. 3982-3992.

[12] OWASP Foundation, "OWASP Top 10:2025," 2025. [Online]. Available: https://owasp.org/Top10/2025/

[13] United Arab Emirates, Federal Decree-Law No. 45 of 2021 Concerning the Protection of Personal Data, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1972

[14] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in Advances in Large Margin Classifiers, A. Smola et al., Eds. Cambridge, MA, USA: MIT Press, 1999, pp. 61-74.

[15] U.S. Environmental Protection Agency, "Documentation chapters for greenhouse gas emission, energy and economic factors used in the Waste Reduction Model (WARM)," WARM version 16, Dec. 2023. [Online]. Available: https://www.epa.gov/warm/documentation-chapters-greenhouse-gas-emission-energy-and-economic-factors-used-waste